USB-C: power + native USB (flashing, logs).

3.3 V: SY8089 buck, Vout = 0.6 V x (1 + 100k / 22k) = 3.33 V

Radar: HLK-LD2410C OUT -> IO4 (wakes the screen).

Scanner on UART0: SparkFun 2D Barcode Scanner (3.3 V)
or a 5 V UART module on J5. Fit only one.

Backlight: TPS61169 boost, I = 204 mV / R13 = 20 mA.
PWM dimming from IO5. 37.5 V overvoltage limit.

Buzzer: MLT-8530 from 5 V, PWM on IO2.

Status LED: side-view SK6812 on the bottom edge, data 3.3 V -> 5 V.

4.3" 800x480 RGB panel, standard 40-pin ribbon.
RGB on GPIOs in connector order (B-G-R 5-6-5 bits, low bits to GND).

GT911 touch ribbon (6-pin, 0.5 mm)

ESP32-S3-WROOM-1-N16R8

J1 USB-C

PWR_FLAG

U2 SY8089AAAC

L1 2.2uH

U1 USBLC6-2SC6

J6 LD2410C radar

J4 SparkFun 2D Barcode Scanner

J5 UART scanner (GM65)

U3 ESP32-S3-WROOM-1-N16R8

SW1 RESET

SW2 BOOT

U5 TPS61169

L2 10uH

D1 B5819W

J2 4.3" RGB panel

J3 GT911 touch

LS1 MLT-8530

D2 1N4148W

Q1 AO3400A

U4 74AHCT1G125

D3 SK6812SIDE-A

H1 M3

H2 M3

H3 M3

H4 M3

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Sheet: /
File: mainboard.kicad_sch

Title: Pantry wall scanner mainboard

Size: A3	Date: 2026-10-27	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 1/1